

## Section 2.5b: Physical Applications - Pumping, Hydrostatic Force

Definition 17.

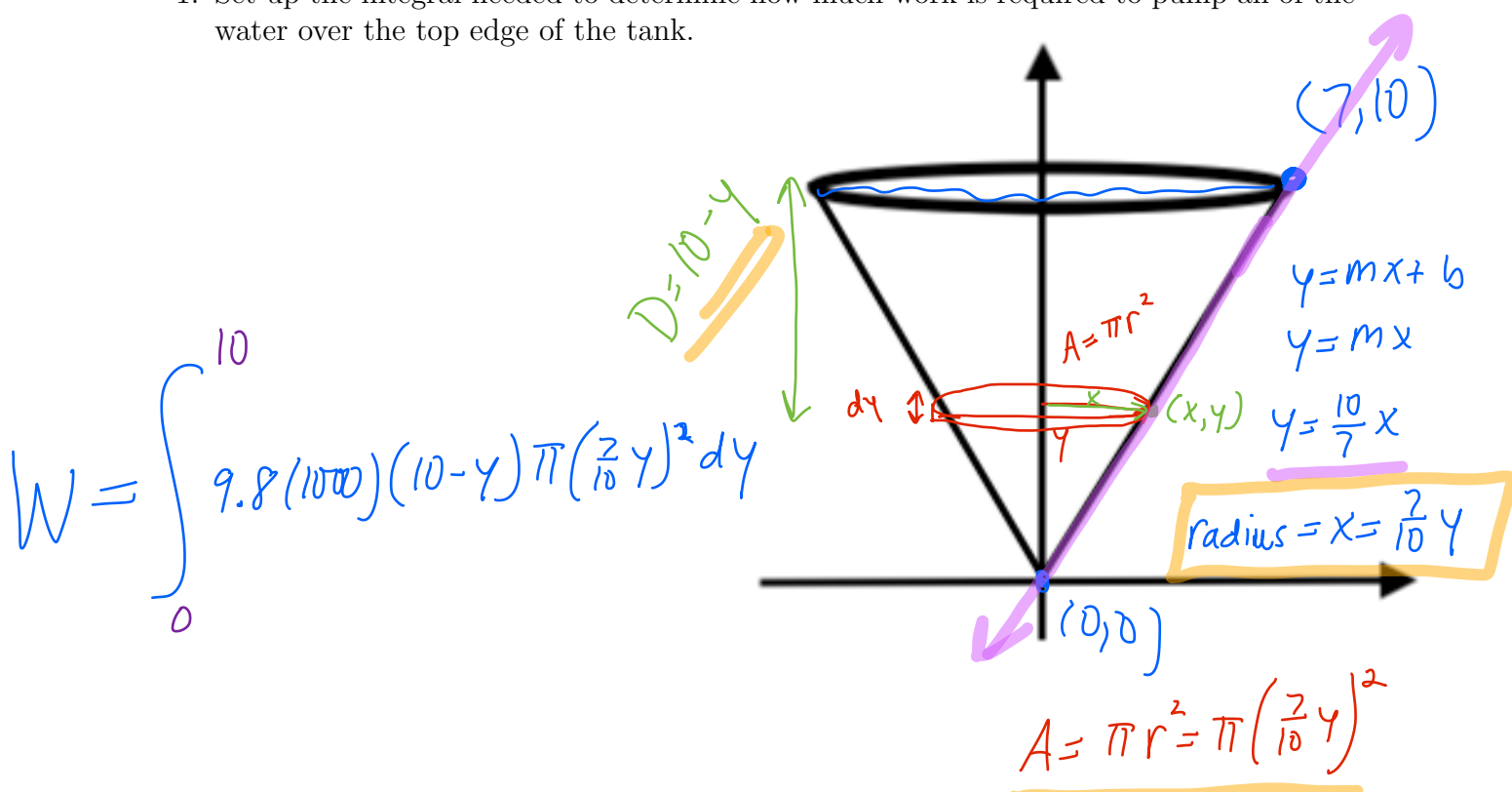
$$\text{Work} = (\text{Force}) \cdot (\text{Distance})$$

Vertical Pumping: "Split into pieces"

$$\sum_{k=1}^n W_k = \sum_k F_k \cdot D_k = \sum_k (\underbrace{\rho}_{m} \cdot \underbrace{V_k}_{a} \cdot g) \cdot y_k \quad \Rightarrow \quad W = \int_a^b \rho \cdot g \cdot V(y) \cdot D(y) \, dy$$

**Example 98.** For a given conical tank, the radius is 7 m and the height is 10 m. Acceleration due to gravity is  $9.8 \frac{m}{sec^2}$ , and the density of water is  $1000 \frac{kg}{m^3}$ .

1. Set up the integral needed to determine how much work is required to pump all of the water over the top edge of the tank.



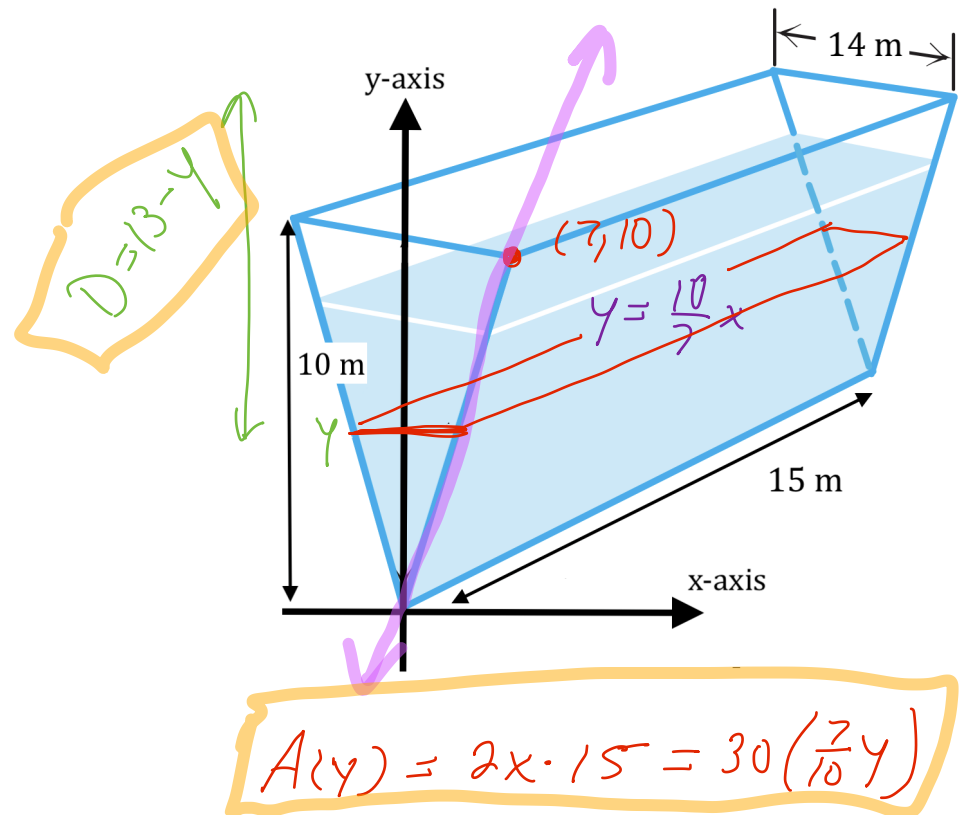
2. Set up the integral needed to determine how much work is required to pump the top 4 m of water out over the top edge of the tank. **Reflect** on modifications needed in the integral set up.

$$W = \int_6^{10} 9.8(1000)(10-y)\pi\left(\frac{7}{10}y\right)^2 dy$$

**Example 99.** For a given olive oil storage tank with triangular ends, the width is  $14\text{ m}$ , the height is  $10\text{ m}$ , and the length is  $15\text{ m}$ . The tank is filled to a depth of  $8\text{ m}$ . Acceleration due to gravity is  $9.8 \frac{\text{m}}{\text{sec}^2}$ , and the density of olive oil is  $900 \frac{\text{kg}}{\text{m}^3}$ .

1. Set up the integral needed to determine how much work is required to pump the top  $4\text{ m}$  of olive oil out through a standpipe that is  $3\text{ m}$  above the top of the tank.

**Reflect** on the modifications (due to geometry) needed in the integral set up from the previous example.

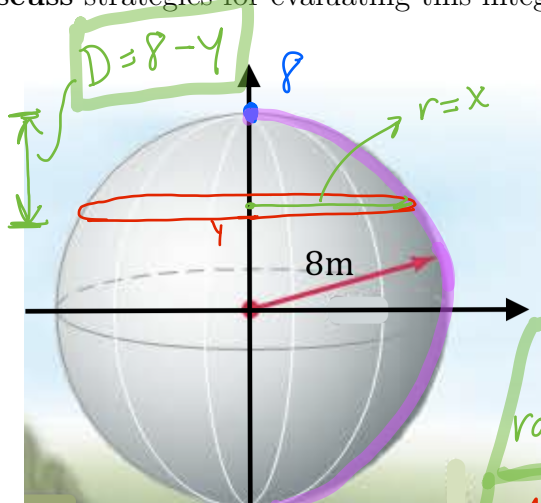


$$W = \int_4^8 9.8(900)(13 - y)30 \left( \frac{7}{10}y \right) dy$$

2. Set up the integral needed to determine how much work is required to pump the olive oil out through a standpipe that is  $3\text{ m}$  above the top of the tank until  $1\text{ m}$  depth of olive oil remains. **Reflect** on changes needed in the integral structure.

**Example 100.** A spherical tank with a radius of  $8\text{ m}$  is completely full of sea water. The sea water is being pumped out the the very top of the tank. Acceleration due to gravity is  $9.8 \frac{\text{m}}{\text{sec}^2}$ , and the density of (salty) sea water is  $1030 \frac{\text{kg}}{\text{m}^3}$ . Set up the integral needed to determine how much work is required to pump all the sea water out through the very top.

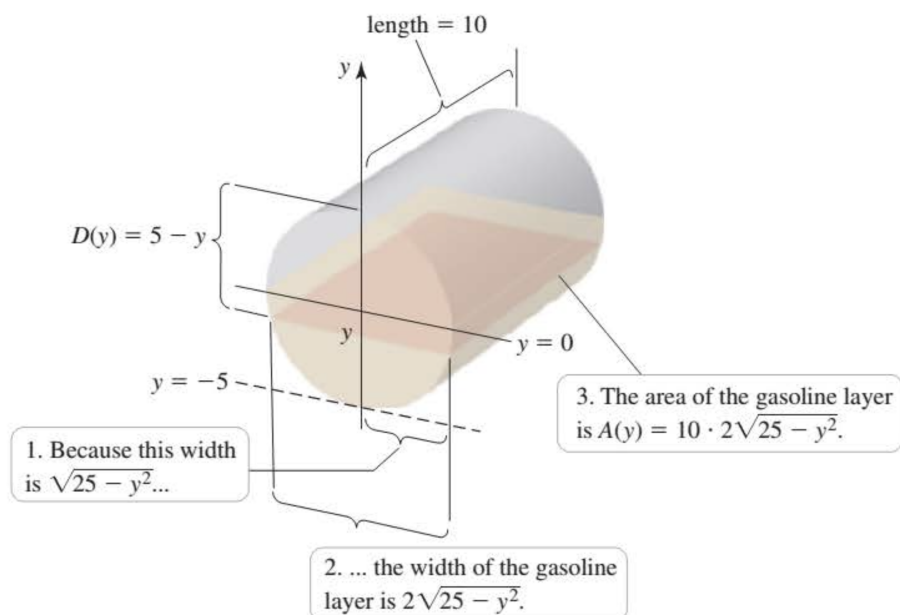
**Discuss** strategies for evaluating this integral.



$x^2 + y^2 = r^2$   
 $x^2 = r^2 - y^2$   
 $x = \pm \sqrt{r^2 - y^2}$   
 $x = \sqrt{r^2 - y^2}$   
 radius  
 $A = \pi (\sqrt{64 - y^2})^2$

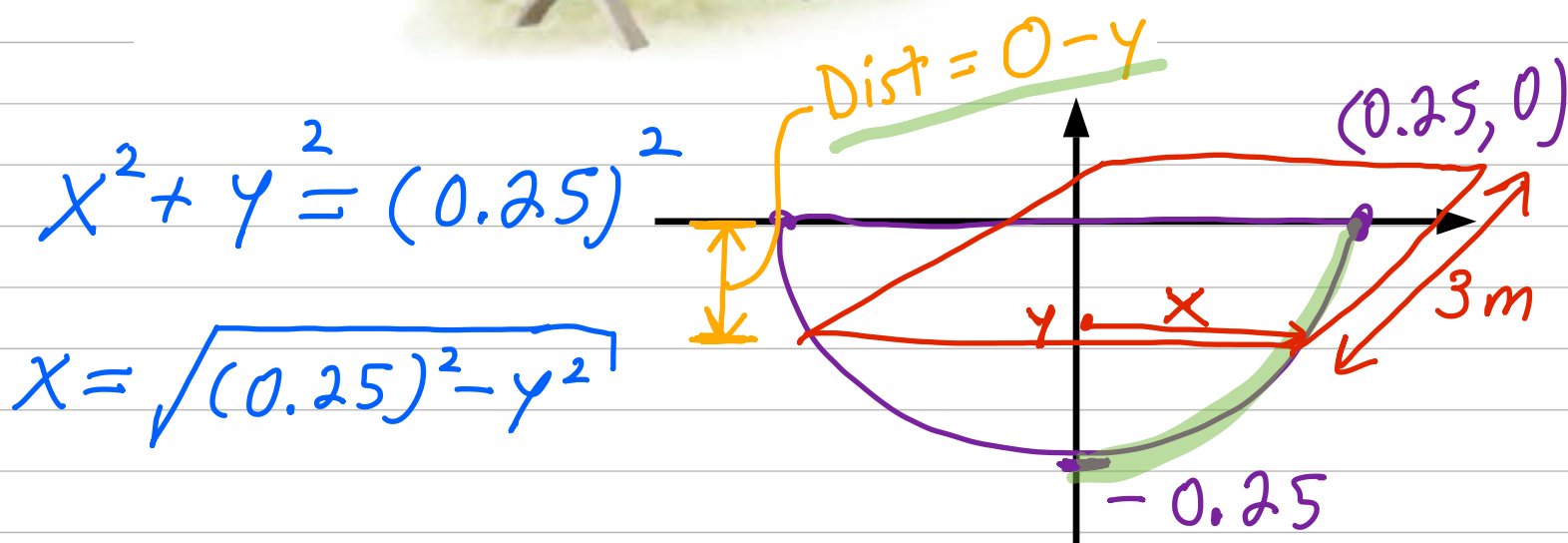
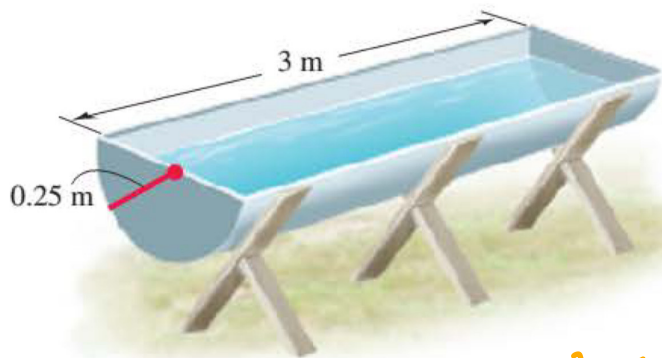
$$W = \int_{-8}^8 9.8(1030)(8 - y) \pi (\sqrt{64 - y^2})^2 dy$$

**Example 101.** Pumping gasoline out of a cylindrical tank on its side with a length of  $10\text{ m}$  and a radius of  $5\text{ m}$  half full of gasoline. **Discuss** the integral set up for the work require to empty the tank through an outlet pipe at the top of the tank. Density of gasoline is  $\rho = 737 \text{ kg/m}^3$ .



42. **Emptying a water trough** A water trough has a semicircular cross section with a radius of 0.25 m and a length of 3 m (see figure).

- How much work is required to pump the water out of the trough (to the level of the top of the trough) when it is full?
- If the length is doubled, is the required work doubled? Explain.
- If the radius is doubled, is the required work doubled? Explain.



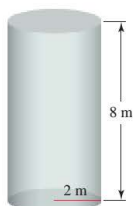
$$A = w \cdot l = 2x \cdot 3$$

$$A(y) = 6\sqrt{(0.25)^2 - y^2}$$

$$W = \int_{-0.25}^0 9.8(1000)(-y) 6\sqrt{(0.25)^2 - y^2} dy$$

15.  $\rho(x) = 2 - \frac{x}{2}$ , for  $0 \leq x \leq 2$
16.  $\rho(x) = 1 + 3 \sin x$ , for  $0 \leq x \leq \pi$
17.  $\rho(x) = x\sqrt{2 - x^2}$ , for  $0 \leq x \leq 1$
18.  $\rho(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 2 \\ 2 & \text{if } 2 < x \leq 3 \end{cases}$
19.  $\rho(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 2 \\ 1 + x & \text{if } 2 < x \leq 4 \end{cases}$
20.  $\rho(x) = \begin{cases} x^2 & \text{if } 0 \leq x \leq 1 \\ x(2 - x) & \text{if } 1 < x \leq 2 \end{cases}$
21. **Work from force** How much work is required to move an object from  $x = 0$  to  $x = 3$  (measured in meters) in the presence of a force (in N) given by  $F(x) = 2x$  acting along the  $x$ -axis?
22. **Work from force** How much work is required to move an object from  $x = 1$  to  $x = 3$  (measured in meters) in the presence of a force (in N) given by  $F(x) = \frac{2}{x^2}$  acting along the  $x$ -axis?
23. **Compressing and stretching a spring** Suppose a force of 30 N is required to stretch and hold a spring 0.2 m from its equilibrium position.
  - a. Assuming the spring obeys Hooke's law, find the spring constant  $k$ .
  - b. How much work is required to compress the spring 0.4 m from its equilibrium position?
  - c. How much work is required to stretch the spring 0.3 m from its equilibrium position?
  - d. How much additional work is required to stretch the spring 0.2 m if it has already been stretched 0.2 m from its equilibrium position?
24. **Compressing and stretching a spring** Suppose a force of 15 N is required to stretch and hold a spring 0.25 m from its equilibrium position.
  - a. Assuming the spring obeys Hooke's law, find the spring constant  $k$ .
  - b. How much work is required to compress the spring 0.2 m from its equilibrium position?
  - c. How much additional work is required to stretch the spring 0.3 m if it has already been stretched 0.25 m from its equilibrium position?
25. **Work done by a spring** A spring on a horizontal surface can be stretched and held 0.5 m from its equilibrium position with a force of 50 N.
  - a. How much work is done in stretching the spring 1.5 m from its equilibrium position?
  - b. How much work is done in compressing the spring 0.5 m from its equilibrium position?
26. **Shock absorber** A heavy-duty shock absorber is compressed 2 cm from its equilibrium position by a mass of 500 kg. How much work is required to compress the shock absorber 4 cm from its equilibrium position? (A mass of 500 kg exerts a force (in newtons) of 500g, where  $g = 9.8 \text{ m/s}^2$ .)
27. **Calculating work for different springs** Calculate the work required to stretch the following springs 0.5 m from their equilibrium positions. Assume Hooke's law is obeyed.
  - a. A spring that requires a force of 50 N to be stretched 0.2 m from its equilibrium position
  - b. A spring that requires 50 J of work to be stretched 0.2 m from its equilibrium position
28. **Calculating work for different springs** Calculate the work required to stretch the following springs 0.4 m from their equilibrium positions. Assume Hooke's law is obeyed.
  - a. A spring that requires a force of 50 N to be stretched 0.1 m from its equilibrium position
  - b. A spring that requires 2 J of work to be stretched 0.1 m from its equilibrium position
29. **Calculating work for different springs** Calculate the work required to stretch the following springs 1.25 m from their equilibrium positions. Assume Hooke's law is obeyed.
  - a. A spring that requires 100 J of work to be stretched 0.5 m from its equilibrium position
  - b. A spring that requires a force of 250 N to be stretched 0.5 m from its equilibrium position
30. **Work function** A spring has a restoring force given by  $F(x) = 25x$ . Let  $W(x)$  be the work required to stretch the spring from its equilibrium position ( $x = 0$ ) to a variable distance  $x$ . Find and graph the work function. Compare the work required to stretch the spring  $x$  units from equilibrium to the work required to compress the spring  $x$  units from equilibrium.
31. **Winding a chain** A 30-m-long chain hangs vertically from a cylinder attached to a winch. Assume there is no friction in the system and the chain has a density of 5 kg/m.
  - a. How much work is required to wind the entire chain onto the cylinder using the winch?
  - b. How much work is required to wind the chain onto the cylinder if a 50-kg block is attached to the end of the chain?
32. **Coiling a rope** A 60-m-long, 9.4-mm-diameter rope hangs freely from a ledge. The density of the rope is 55 g/m. How much work is needed to pull the entire rope to the ledge?
33. **Winding part of a chain** A 20-m-long, 50-kg chain hangs vertically from a cylinder attached to a winch. How much work is needed to wind the upper half of the chain onto the winch?
34. **Leaky bucket** A 1-kg bucket resting on the ground contains 3 kg of water. How much work is required to raise the bucket vertically a distance of 10 m if water leaks out of the bucket at a constant rate of  $\frac{1}{5} \text{ kg/m}$ ? Assume the weight of the rope used to raise the bucket is negligible. (*Hint:* Use the definition of work,  $W = \int_a^b F(y) dy$ , where  $F$  is the variable force required to lift an object along a vertical line from  $y = a$  to  $y = b$ .)
35. **Emptying a swimming pool** A swimming pool has the shape of a box with a base that measures 25 m by 15 m and a uniform depth of 2.5 m. How much work is required to pump the water out of the pool (to the level of the top of the pool) when it is full?
36. **Emptying a cylindrical tank** A cylindrical water tank has height 8 m and radius 2 m (see figure).
  - a. If the tank is full of water, how much work is required to pump the water to the level of the top of the tank and out of the tank?

- b. Is it true that it takes half as much work to pump the water out of the tank when it is half full as when it is full? Explain.

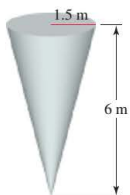


37. **Emptying a half-full cylindrical tank** Suppose the water tank in Exercise 36 is half full of water. Determine the work required to empty the tank by pumping the water to a level 2 m above the top of the tank.

38. **Emptying a partially filled swimming pool** If the water in the swimming pool in Exercise 35 is 2 m deep, then how much work is required to pump all the water to a level 3 m above the bottom of the pool?

39. **Emptying a conical tank** A water tank is shaped like an inverted cone with height 6 m and base radius 1.5 m (see figure).

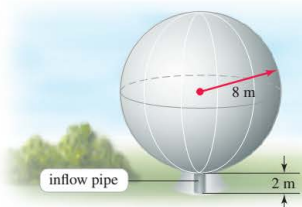
- a. If the tank is full, how much work is required to pump the water to the level of the top of the tank and out of the tank?  
b. Is it true that it takes half as much work to pump the water out of the tank when it is filled to half its depth as when it is full? Explain.



40. **Upper and lower half** A cylinder with height 8 m and radius 3 m is filled with water and must be emptied through an outlet pipe 2 m above the top of the cylinder.

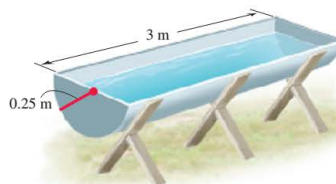
- a. Compute the work required to empty the water in the top half of the tank.  
b. Compute the work required to empty the (equal amount of) water in the lower half of the tank.  
c. Interpret the results of parts (a) and (b).

41. **Filling a spherical tank** A spherical water tank with an inner radius of 8 m has its lowest point 2 m above the ground. It is filled by a pipe that feeds the tank at its lowest point (see figure). Neglecting the volume of the inflow pipe, how much work is required to fill the tank if it is initially empty?



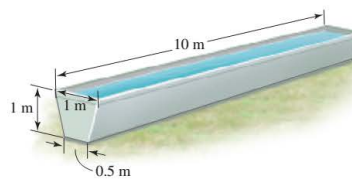
42. **Emptying a water trough** A water trough has a semicircular cross section with a radius of 0.25 m and a length of 3 m (see figure).

- a. How much work is required to pump the water out of the trough (to the level of the top of the trough) when it is full?  
b. If the length is doubled, is the required work doubled? Explain.  
c. If the radius is doubled, is the required work doubled? Explain.



43. **Emptying a water trough** A cattle trough has a trapezoidal cross section with a height of 1 m and horizontal sides of length  $\frac{1}{2}$  m and 1 m. Assume the length of the trough is 10 m (see figure).

- a. How much work is required to pump the water out of the trough (to the level of the top of the trough) when it is full?  
b. If the length is doubled, is the required work doubled? Explain.



44. **Pumping water** Suppose the tank in Example 5 is full of water (rather than half full of gasoline). Determine the work required to pump all the water to an outlet pipe 15 m above the bottom of the tank.

45. **Emptying a conical tank** An inverted cone (base above the vertex) is 2 m high and has a base radius of  $\frac{1}{2}$  m. If the tank is full, how much work is required to pump the water to a level 1 m above the top of the tank?



## Section 3.7: Improper Integrals

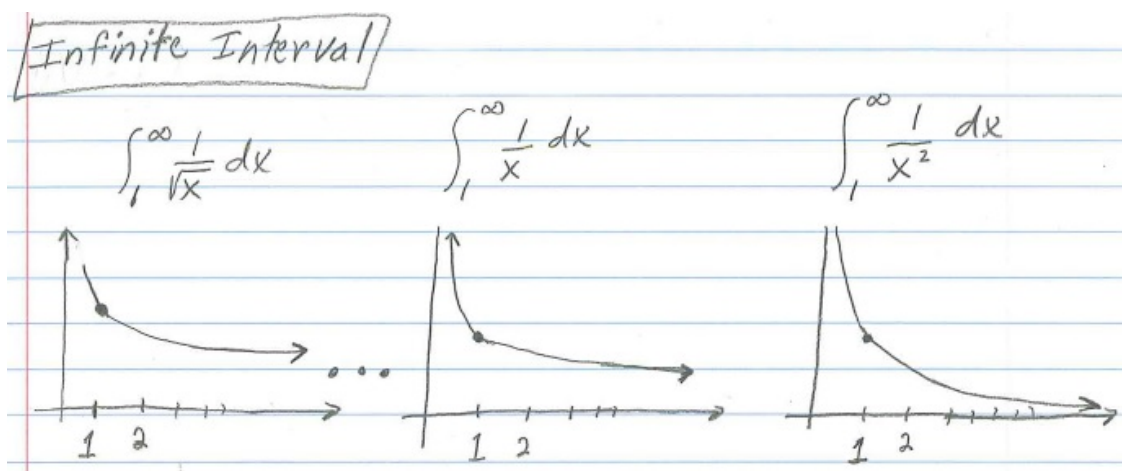
**Definition 18.** An integral is *improper* if either ...

**Type I.** A bound of integration is infinite, e.g.

$$\int_1^{\infty} \frac{1}{x^2} dx$$

**Type II.** The function is undefined (or “infinite”) at an endpoint, e.g.

$$\int_0^1 \frac{1}{x} dx$$



**Definition 19.** For a continuous function  $f$ ,

$$\int_a^{\infty} f(x) dx = \lim_{N \rightarrow \infty} \int_a^N f(x) dx.$$

The integral **converges** if the limit converges to a finite value. Otherwise, the integral is **divergent**.

**Example 106.** Evaluate the improper integral to determine if it converges or diverges.

$$\int_1^{\infty} \frac{1}{x^2} dx \Rightarrow \int_1^N x^{-2} dx = \frac{x^{-1}}{-1} \Big|_1^N = -\frac{1}{N} - \left(-\frac{1}{1}\right) = 1 - \frac{1}{N}$$

$$\int_1^{\infty} \frac{1}{x^2} dx = \lim_{N \rightarrow \infty} 1 - \frac{1}{N} = 1 - 0 = 1$$

converges  $\equiv$

**Example 107.** Evaluate the improper integral to determine if it converges or diverges.

$$\int_0^{\infty} \frac{2}{(x+1)^3} dx \Rightarrow \int_0^N \frac{2}{(x+1)^3} dx = \int_1^{N+1} 2 \cdot u^{-3} du$$

$$\left. \begin{array}{l} u = x+1 \\ u(0) = 1 \\ u(N) = N+1 \\ du = dx \end{array} \right\} = \left. \frac{2u^{-2}}{-2} \right|_1^{N+1} = \frac{-1}{(N+1)^2} + \frac{1}{1^2}$$

$$\Rightarrow \lim_{N \rightarrow \infty} 1 - \frac{1}{(N+1)^2} = 1 - 0 = 1$$

Converges

**Example 108.** Evaluate the improper integral to determine if it converges or diverges.

$$\int_0^{\infty} \frac{1}{\sqrt{x+1}} dx \Rightarrow \int_0^N \frac{1}{\sqrt{x+1}} dx = \int_1^{N+1} u^{-\frac{1}{2}} du$$

$$\left. \begin{array}{l} u = x+1 \\ u(0) = 1 \\ u(N) = N+1 \\ du = dx \end{array} \right\} = \left. 2u^{\frac{1}{2}} \right|_1^{N+1} = 2\sqrt{N+1} - 2\sqrt{1}$$

$$= 2\sqrt{N+1} - 2$$

$$\int_0^{\infty} \frac{1}{\sqrt{x+1}} dx = \lim_{N \rightarrow \infty} 2\sqrt{N+1} - 2 = \infty - 2$$

=  $\infty$   
Diverges.

Observe how the exponent affects convergence.



**Example 109.** Evaluate the improper integral to determine if it converges or diverges.

$$\int_0^{\infty} e^{-x} dx \Rightarrow \int_0^N e^{-x} dx = -e^{-x} \Big|_0^N = -e^{-N} - (-e^0) \\ = 1 - \frac{1}{e^N}$$

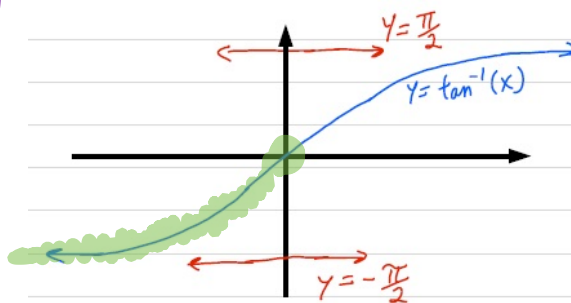
**Example 110.** Evaluate the improper integral to determine if it converges or diverges. If the integral is convergent, keep your answer in exact form in terms of  $\pi$ .

$$\int_{-\infty}^7 \frac{2}{x^2 + 49} dx \Rightarrow \int_{-N}^7 \frac{2}{\underbrace{x^2 + 49}_{7^2}} dx = 2 \cdot \frac{1}{7} \cdot \tan^{-1}\left(\frac{x}{7}\right) \Big|_{-N}^7 \\ = \frac{2}{7} \left[ \underbrace{\tan^{-1}\left(\frac{7}{7}\right)}_{=\frac{\pi}{4}} - \tan^{-1}\left(-\frac{N}{7}\right) \right]$$

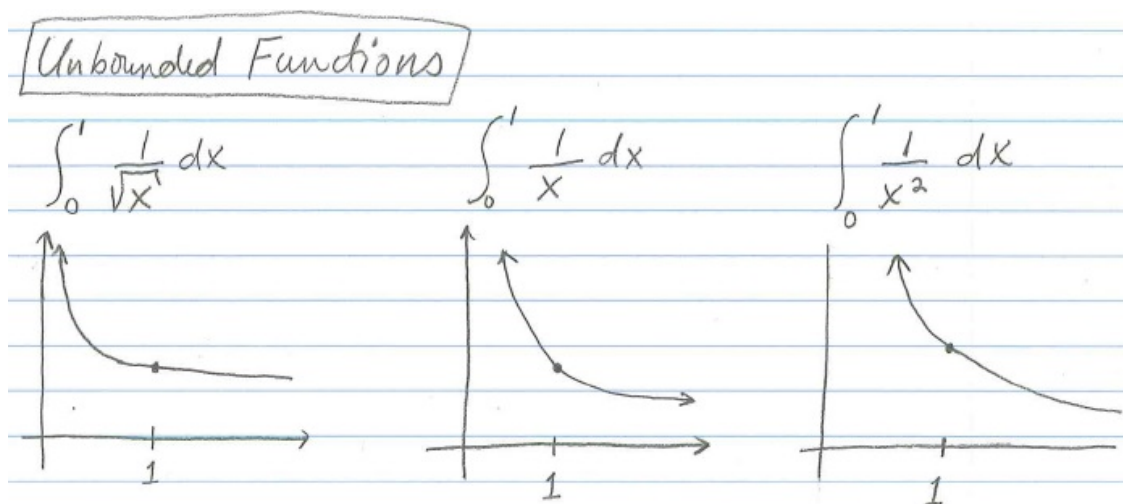
$$\int_{-\infty}^7 \frac{2}{x^2 + 49} dx = \lim_{N \rightarrow \infty} \frac{2}{7} \left[ \frac{\pi}{4} - \underbrace{\tan^{-1}\left(-\frac{N}{7}\right)}_{\rightarrow -\frac{\pi}{2}} \right]$$

$$= \frac{2}{7} \left[ \frac{\pi}{4} - \left(-\frac{\pi}{2}\right) \right]$$

$$= \frac{\pi}{14} + \frac{\pi}{7} \cdot \frac{2}{2} = \frac{3\pi}{14} \text{ Converges.}$$



**Type II.** Improper Integrals.



**Definition 20.** For  $f$  continuous on  $(a, b]$ ,

$$\int_a^b f(x) dx = \lim_{A \rightarrow a^+} \int_A^b f(x) dx$$

**Example 112.** Evaluate the improper integral to determine if it converges or diverges.

$$\int_0^1 \frac{1}{\sqrt{x}} dx \Rightarrow \int_A^1 x^{-1/2} dx = 2x^{1/2} \Big|_A^1 = 2\sqrt{1} - 2\sqrt{A}$$

$$\int_0^1 \frac{1}{\sqrt{x}} dx = \lim_{A \rightarrow 0^+} 2\sqrt{1} - 2\sqrt{A} = 2 - 0 = 2$$

Converges.

**Example 113.** Evaluate the improper integral to determine if it converges or diverges.

$$\int_{-2}^1 \frac{1}{\sqrt{x+2}} dx \Rightarrow \int_A^1 \frac{1}{\sqrt{x+2}} dx = \int_{A+2}^3 u^{-\frac{1}{2}} du$$

$$\left. \begin{array}{l} u = x + 2 \\ u(A) = A + 2 \\ u(1) = 3 \\ du = dx \end{array} \right\}$$

$$= 2u^{\frac{1}{2}} \Big|_{A+2}^3 = 2\sqrt{3} - 2\sqrt{A+2}$$

$$\int_{-2}^1 \frac{1}{\sqrt{x+2}} dx = \lim_{A \rightarrow -2^+} 2\sqrt{3} - 2\sqrt{A+2}$$

$$= 2\sqrt{3} - 2\sqrt{0}$$

**Example 114.** Evaluate the improper integral to determine if it converges or diverges.

$$\int_1^7 \frac{1}{x-1} dx \Rightarrow \int_A^7 \frac{1}{x-1} dx = \ln(x-1) \Big|_A^7$$

$$= \ln(7-1) - \ln(A-1)$$

$$= \ln(6) - \ln(A-1)$$

$$\int_1^7 \frac{1}{x-1} dx = \lim_{A \rightarrow 1^+} \ln(6) - \ln(A-1)$$

$$= \ln(6) - (-\infty) = \infty$$

Diverges. ✓

$$= 2\sqrt{3}$$

Converges. ✓